Agisoft Metashape

Processing Report

31 January 2025

Survey Data

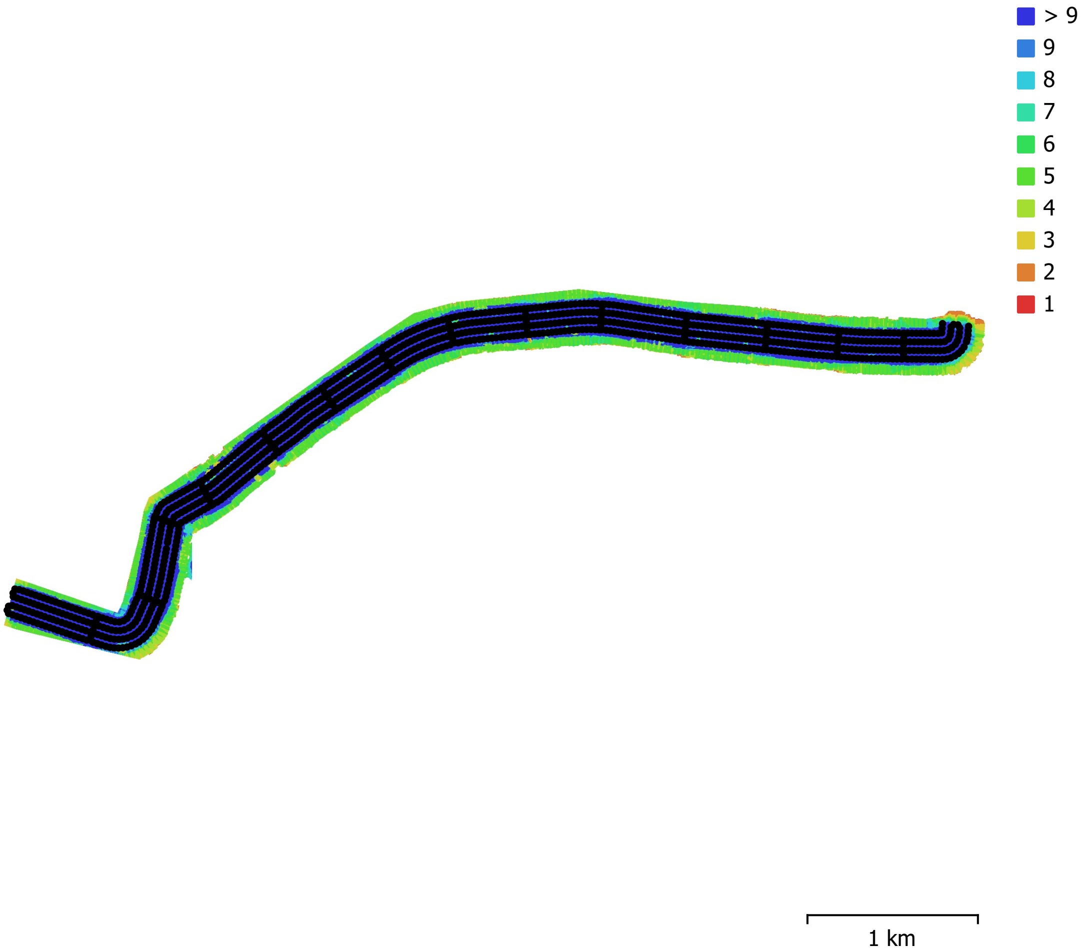

Fig. 1. Camera locations and image overlap.

Number of images:	1,109	Camera stations:	1,109
Flying altitude:	127 m	Tie points:	1,804,938
Ground resolution:	3.27 cm/pix	Projections:	6,646,628
Coverage area:	2.06 km²	Reprojection error:	0.362 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
M3E (12.29mm)	5280 x 3956	12.29 mm	3.36 x 3.36 μm	No

Table 1. Cameras.

Camera Calibration

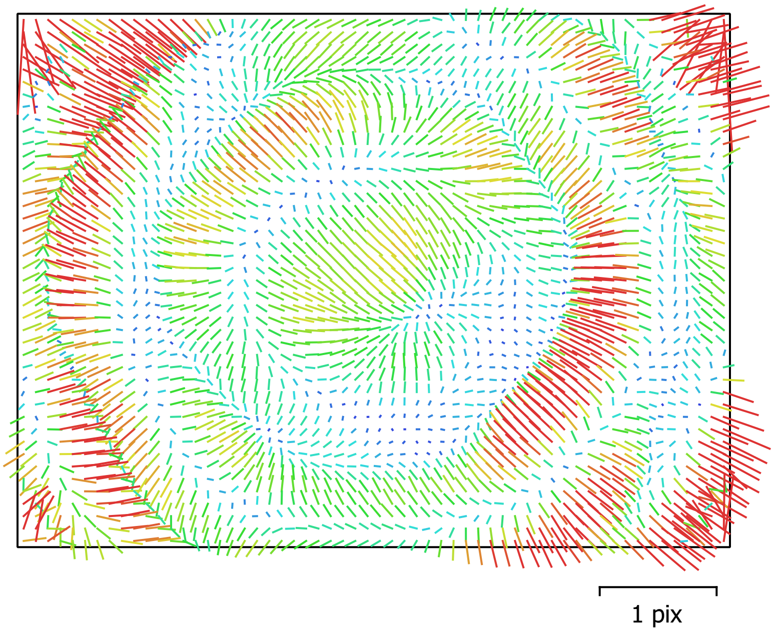

Fig. 2. Image residuals for M3E (12.29mm).

M3E (12.29mm)

1109 images

Type	Resolution	Focal Length	Pixel Size
Frame	**5280 x 3956**	**12.29 mm**	**3.36 x 3.36 μm**

	Value	Error	F	Cx	Cy	B1	B2	K1	K2	K3	K4	P1	P2
F	**3750.77**	0.33	1.00	0.92	-0.85	-0.05	0.10	0.34	-0.58	0.73	-0.78	0.15	-0.09
Cx	**6.82505**	0.016		1.00	-0.78	-0.05	0.10	0.32	-0.54	0.67	-0.72	0.40	-0.09
Cy	**-10.8971**	0.01			1.00	0.04	-0.08	-0.29	0.50	-0.62	0.67	-0.14	0.45
B1	**-0.127833**	0.0017				1.00	-0.00	-0.00	0.02	-0.03	0.04	0.02	0.02
B2	**-0.0599709**	0.0017					1.00	0.04	-0.06	0.08	-0.08	-0.00	0.01
K1	**0.0365423**	1.8e-05						1.00	-0.94	0.84	-0.78	0.05	-0.04
K2	**-0.204729**	0.00012							1.00	-0.97	0.94	-0.08	0.06
K3	**0.462193**	0.00032								1.00	-0.99	0.10	-0.07
K4	**-0.311545**	0.00027									1.00	-0.11	0.08
P1	**0.000274941**	5e-07										1.00	-0.03
P2	**-0.000314009**	4.6e-07											1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

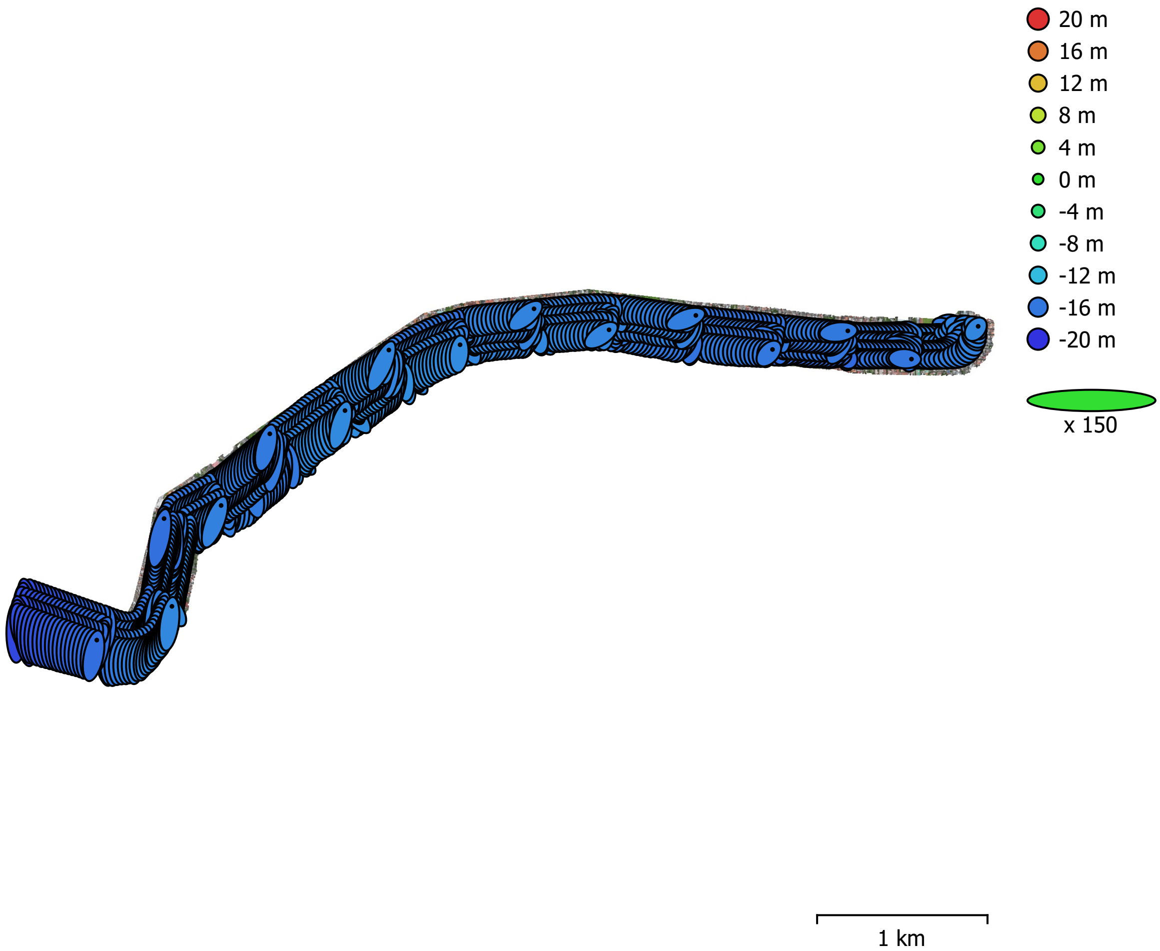

Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
0.222296	0.785242	15.7485	0.816101	15.7696

Table 3. Average camera location error.
X - Easting, Y - Northing, Z - Altitude.

Ground Control Points

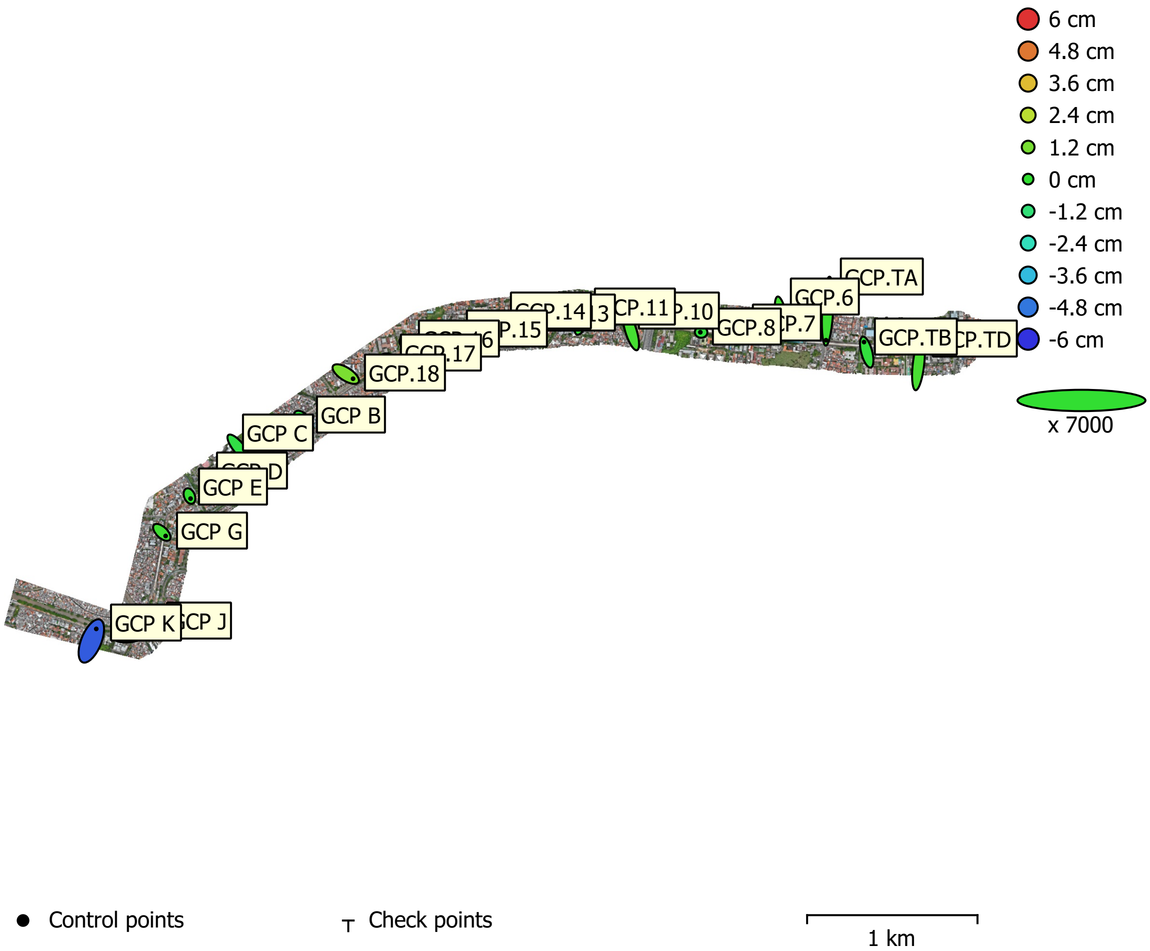

Fig. 4. GCP locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.

Estimated GCP locations are marked with a dot or crossing.

Count	X error (cm)	Y error (cm)	Z error (cm)	XY error (cm)	Total (cm)
21	0.757271	1.73756	1.66717	1.89541	2.52429

Table 4. Control points RMSE.

X - Easting, Y - Northing, Z - Altitude.

Label	X error (cm)	Y error (cm)	Z error (cm)	Total (cm)	Image (pix)
GCP.TD	0.386255	3.39871	0.402948	3.44424	1.808 (8)
GCP.TB	-0.449068	1.69057	-0.198809	1.76046	2.026 (3)
GCP.TA	-0.266684	-4.86269	0.176532	4.87319	2.243 (8)
GCP.6	0.494374	-2.79129	0.291775	2.84971	1.087 (12)
GCP.7	0.219172	-0.662668	-1.0471	1.2584	0.850 (12)
GCP.8	-0.132017	0.00542963	-0.197235	0.237402	0.484 (8)
GCP.10	-0.726098	2.32642	0.351051	2.46225	1.649 (6)
GCP.11	0.495188	1.33553	-0.0441373	1.42507	1.045 (5)
GCP.13	-0.243711	1.05997	-0.392268	1.1562	0.838 (6)
GCP.14	-0.391741	-0.357335	-0.378336	0.651373	0.872 (5)
GCP.15	-0.598814	0.584765	-0.144208	0.849308	0.884 (4)
GCP.16	0.272744	-0.212972	0.223547	0.411971	0.548 (5)
GCP.17	-0.355194	0.0541109	0.763363	0.843691	1.537 (4)
GCP.18	1.22353	-0.782931	1.26587	1.92677	2.059 (7)
GCP B	-0.537096	0.494875	-0.313679	0.794838	0.508 (7)
GCP C	1.11254	-1.37427	-0.347182	1.80192	1.500 (5)
GCP D	0.368563	-0.646358	-0.0341165	0.744836	1.385 (2)
GCP E	0.18059	-0.425941	0.0276478	0.463468	1.616 (1)
GCP G	0.64003	-0.607595	0.233384	0.912841	0.836 (6)
GCP J	-2.34122	-0.691694	5.10598	5.65957	3.094 (13)
GCP K	0.826874	2.0213	-5.28117	5.71491	3.321 (13)
Total	**0.757271**	**1.73756**	**1.66717**	**2.52429**	**1.829**

Table 5. Control points.

X - Easting, Y - Northing, Z - Altitude.

Digital Elevation Model

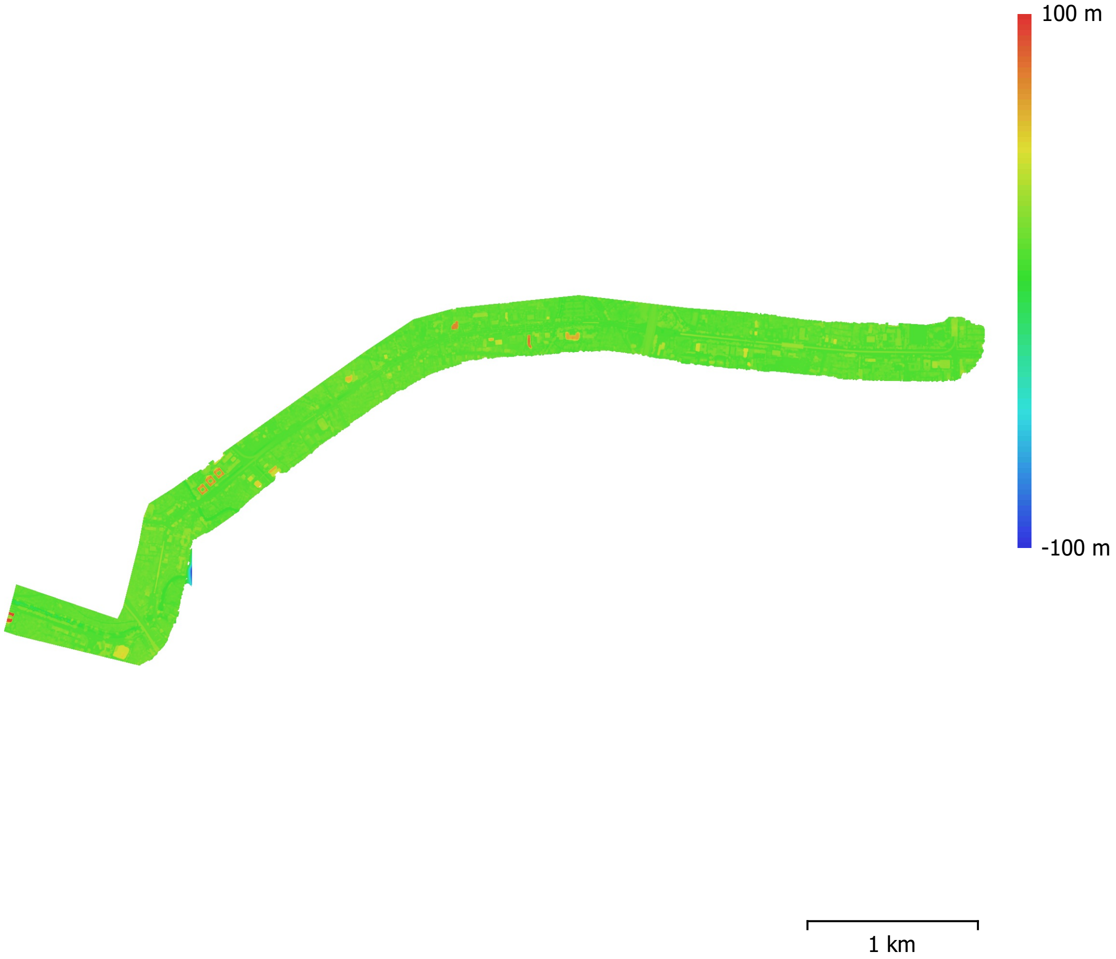

Fig. 5. Reconstructed digital elevation model.

Resolution:	13.1 cm/pix
Point density:	58.5 points/m²

Processing Parameters

General	
Cameras	1109
Aligned cameras	1109
Markers	21
Shapes	
Polygon	1
Coordinate system	WGS 84 / UTM zone 48S (EPSG::32748)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	1,804,938 of 5,368,701
RMS reprojection error	0.158261 (0.362139 pix)
Max reprojection error	1.64288 (5.55384 pix)
Mean key point size	2.2665 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	3.22529
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	60,000
Key point limit per Mpx	1,000
Tie point limit	0
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	No
Matching time	36 minutes 59 seconds
Matching memory usage	9.21 GB
Alignment time	21 minutes 43 seconds
Alignment memory usage	1.67 GB
Optimization parameters	
Parameters	f, b1, b2, cx, cy, k1-k4, p1, p2
Adaptive camera model fitting	No
Optimization time	16 seconds
Date created	2025:01:23 07:04:03
Software version	2.0.2.16102
File size	321.03 MB
Depth Maps	
Count	1109
Depth maps generation parameters	
Quality	Medium
Filtering mode	Moderate
Max neighbors	16
Processing time	2 hours 51 minutes
Memory usage	2.03 GB
Date created	2025:01:23 12:31:57
Software version	2.0.2.16102
File size	2.30 GB
Point Cloud	
Points	208,088,998

Point attributes	
Color	3 bands, uint8
Normal	
Confidence	[illegible] - [illegible]
Point classes	
Created (never classified)	208,088,998
Depth maps generation parameters	
Quality	Medium
Filtering mode	Moderate
Max neighbors	16
Processing time	2 hours 51 minutes
Memory usage	2.03 GB
Point cloud generation parameters	
Processing time	32 minutes 42 seconds
Memory usage	5.06 GB
Date created	2025:01:23 13:04:40
Software version	2.0.2.16102
File size	2.92 GB
Model	
Faces	13,638,421
Vertices	6,836,968
Vertex colors	3 bands, uint8
Texture	4,096 x 4,096, 4 bands, uint8
Depth maps generation parameters	
Quality	Medium
Filtering mode	Moderate
Max neighbors	16
Processing time	2 hours 51 minutes
Memory usage	2.03 GB
Point cloud generation parameters	
Processing time	32 minutes 42 seconds
Memory usage	5.06 GB
Reconstruction parameters	
Surface type	Height field
Source data	Point cloud
Interpolation	Enabled
Strict volumetric masks	No
Processing time	13 minutes 32 seconds
Memory usage	8.42 GB
Texturing parameters	
Mapping mode	Adaptive orthophoto
Blending mode	Mosaic
Texture size	4,096
Enable hole filling	Yes
Enable ghosting filter	Yes
UV mapping time	1 minutes 54 seconds
UV mapping memory usage	2.96 GB
Blending time	1 hours 59 minutes
Blending memory usage	4.97 GB
Blending GPU memory usage	1.50 GB
Date created	2025:01:23 13:07:49
Software version	2.0.2.16102
File size	609.29 MB
DEM	
Size	44,752 x 16,888
Coordinate system	WGS 84 / UTM zone 48S (EPSG::32748)

Reconstruction parameters	
Source data	Point cloud
Interpolation	Enabled
Processing time	3 minutes 6 seconds
Memory usage	318.33 MB
Date created	2025:01:23 15:44:37
Software version	2.0.2.16102
File size	645.05 MB
Orthomosaic	
Size	179,163 x 67,832
Coordinate system	WGS 84 / UTM zone 48S (EPSG::32748)
Colors	3 bands, uint8
Reconstruction parameters	
Blending mode	Mosaic
Surface	Mesh
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	1 hours 52 minutes
Memory usage	9.10 GB
Date created	2025:01:23 16:18:47
Software version	2.0.2.16102
File size	25.04 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.0.2 build 16102
OS	Windows 64 bit
RAM	15.41 GB
CPU	AMD Ryzen 7 5800HS with Radeon Graphics
GPU(s)	AMD Radeon(TM) Graphics (gfx90c)